

Introduction to quantitative time diary analysis

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Plan

- Day 1**
 - 1. Origins and milestones of time diary research
 - 2. Structure and design of time diary data
 - 3. Computing simple time use estimates
- Day 2**
 - 1. Tempograms
 - 2. Work schedules
 - 3. Data quality and weighting
 - 4. Multivariate modelling

Thursday 3 November 2022

- Session 1 2-2.45
 - Origins and milestones of time diary analysis; structure and design of time diary surveys
 - Questions and answers, break
- Session 2 3-3.45
 - Structure and design of time diary surveys (continued).
 - Estimating duration and participation: day- and person-level aggregate statistics
 - Q&A, break
- Session 3 4-5
 - Estimating duration and participation (continued)
 - Q&A; discussion of participants research ideas and interest

How is time studied in sociology?

1. Time as an intrinsic object of study
 - Social theory: ie acceleration (Rosa 2015)
 - Social times and rhythm (Boulin, Tabbonni)
 - Rationalisation and modernity: criticisms of the 'clock time' (Southerton)
2. ... As a empirical tool to investigate social issues
 - Productivity (Organisation studies)
 - Inequality ie Time poverty/exploitation
 - Division of labour within the household
3. Long limited by the availability of reliable, systematic empirical data

Growing interest from the 20th Century onwards

- Early 20th Century: peasant households in Russia & the Fabians (Women in London)
- Soviet economists: Time Budget of Russian Workers from 1922–1924 (Strumilin);
- US Department for Agriculture time use in farms, towns, and later elite educated “college”, women
- In the UK: Mass Observation from 1937 and the BBC listeners surveys
- Growing acknowledgement of the importance of measuring time in modern societies

What drove this interest for time use?

- Monitoring economic productivity, in particular household productivity
- Labour force / workplace behaviour
- Social (central) planning
- Better understand consumer behaviour
- Investigate social issues ie poverty, gender inequality
- More broadly part of continued move towards greater rationalisation of modern societies (Weber)

Contemporary time studies

- Large scale mostly Western comparative datasets
 - Alexander Szalai's *The Use of Time*
 - 12 countries, mostly urban households
 - First systematic time diary data collection: '*who does what, where, with whom, over 24h*'
 - Pioneering, remarkable cooperation for the times
 - Multinational Time Use Study (MTUS)
 - J Gershuny at CTUR, UCL
 - Main source of international harmonised time use data
 - Large archive: 55 years, 1 million diaries, 30 countries, 70 surveys
 - EUROSTAT's Harmonised European Time Use Survey
- A large number of national studies ie American Time Use Survey; Indian Time Use Survey, etc..

What does a typical Time Use survey look like?

- Example: 2015 UK Time Use Survey
 - 16,550 diary days of 10,208 respondents in 4,238 households in the UK
 - Diaries for respondents aged 8 and above
 - Collected all year round
- Two main components:
 1. Comprehensive individual survey ie
 - Socio-demographic characteristics, health, etc..
 2. Time diary
 - 10 minutes time slots
 - Two days per person (one weekday/one day at the weekend)
- Units of observation are *days* rather than persons

What is recorded in time diaries?

- **Activities:** main and secondary ie 'eating dinner and watching TV'
- **Location:** where the activities took place (ie at home, at work, etc)
- **Copresence:** who else was present at that time
- More recently:
 - Whether a device (ie phone, computer, tablet) was used
 - Immediate well-being: enjoyment
- Any combination of the above is an *episode*
- Activities recoded using a harmonised nomenclature (HETUS)

A paper time diary

`pics/UKTUS-diary.png`

Common estimates derived from time diaries data

- The time spent on activities ie their duration
- Their sequencing through the day
- Whether activities are carried out or not ie their probability.
- The importance of activities relative to other activities or other people
- Their change over time

Durations of activities

(Source: Gershuny et al 2020)

Duration of activities?

- For simplicity, activities were grouped in 4 categories
- For each time point, we have the average amount of time spent on each activity
- Activities all add up to 24h ie 1,440 minutes.
- We can compare time use between groups
- Three main lessons
 - Overall impression of stability over time
 - Little change in sleep and leisure duration
 - Opposite trends in paid and unpaid work
 - Marked gender differences

Sequencing of activities through the day – then...

(Source: Gershuny et al 2020)

Sequencing of activities through the day... And now

(Source: Gershuny et al 2020)

Sequencing of activities

- These 'tempograms' show by time of the day, the proportion of people engaged in given activities
- They provide an overview of daily rhythms:
 - People busy working between 8 and 6PM
 - Leisure activity between 6 and 10 PM
 - Meal times
- ... But also social change
 - The weekdays of men and women have converged over time
 - Changes in leisure behaviour
 - Spread of eating time: erosion of the traditional meal
 - Lifestyles have become more diverse

Summary

- There has been an increase of interest for the way men, women and households spend their time over the 20th century
- Systematic time diary data collection from the 1960s
- There is now large amount of comparative data enabling time use research
- Time diaries record primary, secondary activities and their context
- Stability *and* change in daily behaviour over the last 50 years; gender contrasts

2. Time squeeze and acceleration

1. Are we time-squeezed?
2. Is time accelerating?
3. Empirical evidence

Too many things to do, not enough time?

1. Common trope in the news:
 - *'Too busy to read'* (The Guardian)
 - *Technology is speeding up our sense of time* (The Telegraph)
 - *Give up your time to have more time* (The Telegraph)
 - *Time pressure affects almost every household* (the Financial Times)
2. This rings true, but is this supported by empirical evidence?

Echoed in sociological research...

- The overworked American (Schor 1993)
 - 'Squirrel cage': work and spend cycle
- The time bind (Hochschild 1998)
 - People in the US spend more time on paid work than ever before...
 - ... Which leaves them with not enough time to fulfil their domestic duties
 - And feeling dissatisfied, with some ambivalence
- Abundant empirical literature on work life balance/conflict

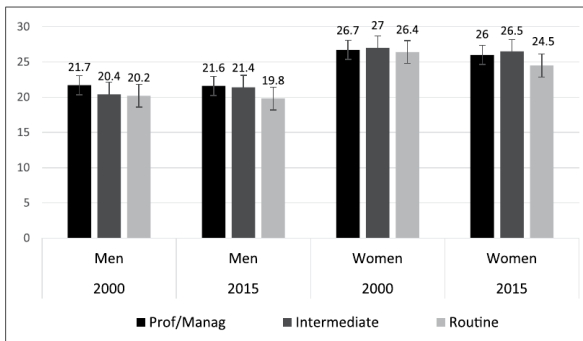
Is time accelerating?

- Acceleration (Rosa 2003)
 - Technological acceleration (communication, transport & production) quest for speed in advanced capitalist societies;
 - Acceleration of social change
 - Acceleration of the pace of life
- Harriedness (Southerton et al 2005)
 - Volume ie duration
 - Fragmentation/disorganisation: work schedules
 - Temporal density of work and leisure

What can we make of this?

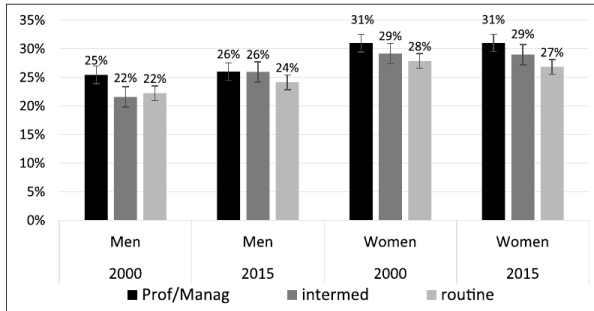
- Little evidence of overall working time increase
 - ... But differences between groups may cancel each other
- Parenting ideals: hyperparenting
- Dual earners, high pressured professional families may have greater public visibility
- 'Busyness' as badge of honour (Gershuny)?

Time fragmentation (Gershuny & Sullivan 2016)



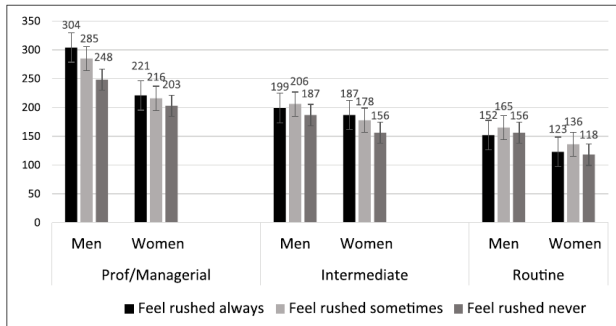
- Number of activities by gender and social class, UK 2015.
- Few differences in the number of activities

... Multitasking (Gershuny & Sullivan 2016)



- % of daily activities multitasked, by gender and social class, UK 2015.
- Differences are small

Device use & stress (Gershuny & Sullivan 2016)



- Number of activities by gender, social class and feeling rushed: UK 2015.
- Professional and managerial who always feel rushed report more time using devices.

Summary

- Report of acceleration of pace of life and scarceness of time
- Matched by an increased sense of being rushed
- Little systematic empirical evidence that overall working time, multitasking and fragmentation have increased
- But use of IT device is associated with feeling rushed.
- Expectation about childcare and leisure may have raised